

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

Name: Prachi Manwal

Roll No: 2501730365

Email: 2501730365@krmu.edu.in

Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def analyze_recursive_iterative(n):
    def rec_factorial(val):
        if val <= 1:
            return 1
        return val * rec_factorial(val - 1)

    rec_fact_val = rec_factorial(n)
    rec_fact_ops = n + 1

    inter_fact_val = 1
    for i in range(1, n + 1):
        iter_fact_val *= i
    iter_fact_ops = n

    fib_calls = 0

    def rec_fibonacci(val):
        nonlocal fib_calls
        fib_calls += 1
        if val == 0:
            return 0
        if val == 1:
            return 1
        return rec_fibonacci(val - 1) + rec_fibonacci(val - 2)

    rec_fib_val = rec_fibonacci(n)
    rec_fib_ops = fib_calls

    if n == 0:
        iter_fib_val = 0
    elif n == 1:
        iter_fib_val = 1
    else:
        a, b = 0, 1
        for _ in range(2, n + 1):
            a, b = b, a + b
        iter_fib_val = b
    iter_fib_ops = n

    return [
        "Computation Analysis Report",
        f"Recursive Factorial: {rec_fact_val}",
        f"Iterative Factorial: {iter_fact_val}",
        f"Recursive Fibonacci: {rec_fib_val}",
        f"Iterative Fibonacci: {iter_fib_val}",
        f"Recursive Factorial operations: {rec_fact_ops}",
        f"Iterative Factorial operations: {iter_fact_ops}",
        f"Recursive Fibonacci operations: {rec_fib_ops}",
        f"Iterative Fibonacci operations: {iter_fib_ops}"
    ]
]
```